

Smart Workout: A RAG-Assisted Decision Support and Business Intelligence System for Personalized Weight Training

Final report

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ABSTRACT

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INTRODUCTION

1.1 Background of the Study

Weight training is widely practiced for strength and fitness, yet many gym users struggle to plan effectively: they are uncertain which exercises suit a given body part, how to adapt to available equipment, how to structure a weekly timetable, and whether their physical condition supports a particular plan. Most decisions rely on generic online routines that ignore personal factors — age, sleep, BMI, physiological indicators, equipment, schedule — producing a mismatch between plan and readiness. This project proposes Smart Workout, a combined Decision Support System (DSS) and Business Intelligence System (BIS) (Power, 2002; Chaudhuri et al., 2011) that unifies multi-source real-world data, a Retrieval-Augmented Generation (RAG) knowledge layer, and supervised ML models within an interactive web dashboard, transforming user-provided inputs into descriptive insight, predictive outcome estimates, and prescriptive training plans while preserving human oversight. The primary target users are gym members seeking personalized training guidance; fitness coaches constitute a secondary audience using the dashboard for decision-support reference.

1.2 Objectives of the Study

The project is expected to deliver a functional DSS/BIS prototype that fulfils the following objectives, explicitly covering the descriptive, predictive, and prescriptive analytics layers required of a BI system:

- **Data collection & preparation:** clean and standardize multi-source public datasets (gym membership and check-in records, exercise catalogue, gym-member workout tracking, sleep & lifestyle, daily nutrition; details in §4) alongside structured user input.
- **Descriptive analytics:** segment users into lifestyle profiles via clustering; present a relational gym membership view (member activity, subscription trends, usage analytics) as an interactive dashboard; and present cross-dataset insights (calorie-burn patterns, exercise coverage, nutrition macro mix).
- **Predictive analytics:** train and compare multiple candidate models per task via cross-validation: (a) calorie-burn-rate regression (kcal/hr); (b) Workout-intensity 3-class clas-

sification (Low/Mid/High).

- **Prescriptive analytics:** generate a daily/weekly training plan via a rule-based engine conditioned on the predicted intensity, readiness, target body part, and available equipment; the predicted calorie-burn rate is displayed alongside the plan. A rule-derived Training Readiness band (computed from user input; rules drawn from the Sleep Health dataset) guides the plan generator (see §5).
- **RAG knowledge retrieval & AI chat assistant:** retrieve exercise instructions, equipment-substitution rules, and scheduling guidance via an interactive chat interface (RAG retrieval is core; LLM chat is a planned extension).
- **BI dashboard:** React-based dashboard (with embedded Tableau views) presenting a relational gym membership view, multi-dataset descriptive analytics, personalized predictions, prescribed plan, RAG-sourced explanation, and embedded chat.

CHAPTER 2

PROBLEM DEFINITION

CHAPTER 3

METHODOLOGY AND SUPERDECISIONS IMPLEMENTATION

3.1 AHP Model Structure

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Final Synthesised Ranking

CHAPTER 5

SENSITIVITY ANALYSIS

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CONCLUSION

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